ANANYA GANDHI

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Education

Indian Institute of Science Education and Research (IISER) - Bhopal **2020 – 2025**
*BS-MS in Electrical Engineering and Computer Science | **CPI: 7.56/10*** *Bhopal, India*

- Major: Electrical Engineering and Computer Science (EECS)
- Minor: Data Science and Engineering (DSE)

MS Thesis: Posing Flocking as a MARL Problem *Aug 2024 – May 2025*
Advisor: Prof. Sujit P. B. *IISER Bhopal*

- **Goal:** Formulated decentralized flocking as a Multi-Agent Reinforcement Learning (MARL) problem using a Partially Observable Markov Game framework.
- **Method:** Designed hybrid reward functions and trained UAV swarm agents using PPO and SAC algorithms within a Unity-based simulator.
- **Outcome:** Demonstrated emergent stable flocking in dynamic, obstacle-rich environments, with PPO + curriculum learning achieving superior performance.
- Code | Final Report | Mid-Year Report | Slides

BS Thesis: Moving Depot (MOD): An Efficient Depot Motion Strategy for Multi-Robot Foraging *Jan 2024 – Apr 2024*
Advisor: Prof. Sujit P. B. *IISER Bhopal*

- **Goal:**Developed a novel moving depot strategy (MOD) that dynamically relocates the depot based on foraging patch quality, using a potential field method for obstacle avoidance.
- **Method:** Conducted agent-based simulations demonstrating up to **50% performance improvements** over stationary depots and multiple-place foraging algorithms in dynamic environments.
- **Outcome:** This work led to an accepted *Extended Abstract* and *Poster Presentation* at the *ANTS 2024 International Conference on Swarm Intelligence.*

Relevant Coursework: Multi-Agent Reinforcement Learning, Artificial Intelligence, Network Science, Control Systems, Data Science ML, Quantum Comp., Computer Vision, Applied Optimization, Probability & Statistics, Real Analysis.

Research Interests

Distributed Control and Learning in Multi-Robot Systems, Swarm Intelligence, Multi-Agent Reinforcement Learning, Theory of Mind in Multi-Agent Systems, Drone Swarming and Simulation, Decentralised Decision-Making under Uncertainty

Honours and Awards

Kanako Miura Travel Award | *IEEE-RAS International Conference on Humanoid Robots 2024* **Nov 2024**
Awarded by the IEEE Robotics and Automation Society (RAS) Technical Committee on Humanoid Robotics. *Nancy, France*

Publications

P. Ingle, A. Gandhi, P.B. Sujit, *"Moving Depot (MOD): An Efficient Depot Motion Strategy for Multi-Robot Foraging". In Proceedings of the 13th International Conference on Swarm Intelligence (ANTS 2024)*, Konstanz, Germany, 2024. (Extended Abstract)

Presentations & Talks

A. Gandhi, *"Adaptive Swarm Strategies and Multi-Agent Reinforcement Learning for Optimization". Lightning Talk, Women in Engineering Luncheon, IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, Nancy, France, Nov 2024. (Recipient of the Kanako Miura Travel Award)

A. Gandhi, P.B. Sujit, *"Multi-Drone Reinforcement Learning for Swarm Formation and Navigation". Poster Presentation, EECS Research Symposium 2025, IISER Bhopal, 2025*

P. Ingle, A. Gandhi, P.B. Sujit, *"Moving Depot (MOD): A Strategy for Efficient Multi-Robot Foraging". Poster Presentation, EECS Research Symposium 2025, IISER Bhopal, 2025.*

A. Gandhi, Peter Dayan, *"Modelling Theory of Mind in Buyer-Seller Interactions". Project Presentation, CASCB Summer School, Konstanz School of Collective Behaviour, 2025.*

Research Experience

Summer School Participant **July 2025**
Konstanz School of Collective Behaviour (KSCB), University of Konstanz *Germany*

- Selected for interdisciplinary program on collective behaviour across robotics, biology and complex systems
- Studied theoretical foundations of collective intelligence: swarm dynamics, coordination mechanisms & emergent behaviours
- Developed and presented a project on *"Modelling Theory of Mind in Buyer–Seller Interactions"* under the mentorship of Prof. Peter Dayan.

Undergraduate Researcher (with Prof. Sujit P. B.) **Aug 2022 – May 2025**
Multi-rObot autONomy (MOON) Lab, IISER Bhopal

Emergent Flocking via Multi-Agent Reinforcement Learning (MS Thesis)

- Formulated decentralized flocking as a MARL problem using a Partially Observable Markov Game framework.
- Trained UAV swarms using PPO and SAC within a Unity-based simulator, demonstrating emergent stable flocking in dynamic, obstacle-rich environments.
- Achieved superior performance with PPO and curriculum learning.

Moving Depot (MOD): Multi-Robot Foraging Strategy (BS Thesis)

- Developed a novel depot motion strategy that dynamically relocates the depot based on foraging patch quality.
- Demonstrated up to **50% performance improvements** over baselines in agent-based simulations.
- This work led to an accepted extended abstract & poster at the *ANTS 2024 International Conference on Swarm Intelligence*

Danger Communication for Multi-Robot Foraging

- Designed a swarm algorithm for a danger-based signal communication system.
- Validated the approach through comprehensive Monte-Carlo simulations.

CS Theory Winter Residency Program Student **Dec 2024**
Topics in Computational Social Choice (with Prof. Rohit Vaish), IIT Delhi

- Explored the mathematical foundations of social choice theory and collective decision-making.
- Analyzed algorithms for fairness, efficiency, and incentives in economic decision-making problems.

Summer School Student **July 2023**
Dynamic Resource Allocation in Networks (with Prof. Alexandre Reiffers-Masson), IISc Bangalore

- Explored mathematical and ML techniques for NP-Hard dynamic resource allocation problems in communication networks.
- Implemented heuristics including LP-based and Whittle index policies; applied theory of Weakly-Coupled MDPs and Restless Multi-Armed Bandits.

Systems Engineering Experience

Indian Army Internship Program (IAIP) 2025, (Selected Candidate) **Jan 2026 – Mar 2026**
Indian Army AI Incubation Centre (IAAIIC) under DGIS, Bengaluru

Private Cloud & MLOps Platform Infrastructure

- Architected and deployed an enterprise-grade MLOps platform across 7 heterogeneous systems (Ubuntu, Fedora, Windows 11 + WSL2) with hybrid Docker/native service architecture
- Designed nginx reverse proxy layer with SSL/TLS termination and host-based routing for 15+ `*.lab.local` HTTPS services using FreeIPA-issued wildcard certificates
- Implemented centralized identity infrastructure with FreeIPA (LDAP + Kerberos + DNS + CA) federated through Keycloak OAuth2/OIDC for SSO across JupyterHub, MLflow, GitLab, Jenkins, MinIO, Mattermost, and MediaWiki
- Deployed ML platform components — JupyterHub (multi-user notebooks), MLflow (experiment tracking), and oauth2-proxy (SSO wrapper) — as native systemd services on a Windows 11 WSL2 host
- Configured S3-compatible object storage (MinIO) with Keycloak OIDC authentication for Docker image distribution, ML dataset management, and cross-system artifact sharing
- Built observability stack with Prometheus metrics aggregation and Grafana dashboards for GPU utilization, active user sessions, and service health monitoring

Technical Skills

Programming Languages: Python, C++, MATLAB, C# (Unity)
ML/AI Frameworks: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, PettingZoo, Xanadu-PennyLane
MLOps & Data Infrastructure: JupyterHub, MLflow, Prometheus, Grafana, MinIO (S3), Docker, Docker Compose
DevOps & Infrastructure: Git, GitLab, Jenkins (CI/CD), Nginx (reverse proxy), PostgreSQL, systemd, SSH, Linux
Identity & Security: Keycloak (OAuth2/OIDC), FreeIPA (LDAP/Kerberos), SSL/TLS, SSO architecture
Collaboration & Documentation: Mattermost, MediaWiki, LaTeX, Jupyter Notebooks
Design & Simulation: Shapr3D, Unity Simulation Environment, Wolfram Mathematica

Course Projects

Collaborative Multi-Car Racing using Multi-Agent Reinforcement Learning | *Python, PettingZoo, Stable-Baselines3* **2024**
- Implemented MADQN to train autonomous agents for cooperative racing in a grid-based track environment

Feature Selection for Few-Shot Mixed-Type Building Classification | *Python, PyTorch, Genetic Algorithms* **2024**
- Applied genetic algorithms and contrastive learning to classify mixed-type buildings from Street-View images
- Achieved superior accuracy compared to ResNet-50 with DINOv2, demonstrating effectiveness of the approach
- Explored few-shot learning to reduce dependency on large labeled datasets for urban scene understanding

Network Analysis of the Indian Stock Market | *Python, NetworkX, Pandas* **2024**
- Conducted graph-based analysis of stock market networks using centrality measures (degree, betweenness, eigenvector)
- Identified Federal Bank and ITC as systematically important nodes based on daily closing price correlations
- Developed visualizations to represent complex financial relationships and market dynamics

Quantum HHL Simulation Project | *Python, Xanadu PennyLane, Wolfram Mathematica* **2023**
- Implemented the Harrow-Hassidim-Lloyd (HHL) algorithm to solve a 2D non-Hermitian linear system by transforming it into Hermitian form
- Designed and simulated quantum circuits using the PennyLane framework, achieving normalized solutions with minor deviations
- Analyzed challenges in quantum algorithm implementation, including approximation errors in controlled rotations

Butterfly/Moth Image Classification (100 Species) | *Python, PyTorch* **2022**
- Fine-tuned a ResNet50 pre-trained model to classify 100 species of butterflies and moths
- Achieved 77.8% validation accuracy and 79.4% test accuracy using transfer learning

Achievements

Indian Army Internship Program (IAIP) 2025 Jan-Mar 2026
*Selected as **1 of 10 successful interns** from **42,323 applicants** (0.002% acceptance rate)* AARIC, Bangalore
through multiple interview rounds for the Indian Army Internship Program under DGIS.

Foundation Level Completion, 2022
Data Science and Statistics Programme, IIT Madras

Maharashtra State Board Junior College Topper - Computer Science 2020
Jr. College Topper in CS –Achieved 97.5% in Computer Science; 92% in 10th boards (2018), 87% in 12th boards(2020). Pune, India

INSEF Regional Fair - Special Mentions Award Nov 2017
One of 21 students selected for the final round from over 2,000 participants. IISER Pune
Recognized for chemistry research project "Element Flame Colours: A Trend in the Periodic Table?"

Infosys Spark: Catch Them Young (CTY) Program Selectee July 2016
Selected as one of 40 students from over 3,000 top-performing students for an IT workshop Infosys, Pune

Leadership and Extracurricular Activities

Arbitrage: The Quantitative Finance Club of IISER Bhopal **Founding Date – Jan 2025**
Founder & Coordinator *IISER Bhopal*
- Established the first quantitative finance-focused club across all IISERs
- Developed educational programs to foster financial literacy and quantitative skills among peers

IISERB Book Club **Aug 2022 – May 2025**
Coordinator (2022–2023), Advisor (2023–2025) *IISER Bhopal*
- Revitalized the club post-pandemic, leading recruitment and management of the core committee
- Expanded the club's reach and impact, earning recognition from the Education Minister of Madhya Pradesh
- Organized literary events, author talks, and reading sessions to foster intellectual community

Additional Information

Languages: English (Fluent), Hindi (Fluent), Marathi (Native), Spanish (Beginner)
Hobbies: Literature, Travelling, Hiking, Letter Writing & Philately, Chess, Basketball
Workshops: Energy Conversion and Storage, 3D Design and Printing, IPR Workshop